



# Common Ports & Protocols

Use this when: you read a firewall rule, scan or netstat output and need the service behind a port.

CCNA · Network+ · A+ Core 1  
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## Port map: 32 ports to know

Port · protocol · transport

|                            |                     |                   |               |                     |                         |                     |                  |                     |
|----------------------------|---------------------|-------------------|---------------|---------------------|-------------------------|---------------------|------------------|---------------------|
| <b>Remote &amp; files</b>  | 20/21 FTP TCP       | 22 SSH TCP        | 23 Telnet TCP | 69 TFTP UDP         | 137-139 NetBIOS UDP/TCP | 445 SMB/CIFS TCP    | 3389 RDP TCP/UDP | 5900 VNC TCP        |
| <b>Web &amp; email</b>     | 80 HTTP TCP         | 443 HTTPS TCP/UDP | 25 SMTP TCP   | 587 Mail submit TCP | 110 POP3 TCP            | 995 POP3S TCP       | 143 IMAP TCP     | 993 IMAPS TCP       |
| <b>Network services</b>    | 53 DNS UDP/TCP      | 67/68 DHCP UDP    | 123 NTP UDP   | 161/162 SNMP UDP    | 514 Syslog UDP          | 6514 Syslog-TLS TCP | 5060 SIP UDP/TCP | 5061 SIP-TLS TCP    |
| <b>Identity &amp; data</b> | 88 Kerberos UDP/TCP | 389 LDAP TCP/UDP  | 636 LDAPS TCP | 49 TACACS+ TCP      | 1812/13 RADIUS UDP      | 1433 MS SQL TCP     | 3306 MySQL TCP   | 5432 PostgreSQL TCP |

🔒 Encrypted (TLS / SSH)
 🔓 Cleartext (readable on the wire)
 ⚙️ Depends on config

## Swap cleartext for secure

|                     |   |               |                        |
|---------------------|---|---------------|------------------------|
| Telnet 23           | → | SSH 22        | encrypted              |
| FTP 21 · TFTP 69    | → | SFTP / SCP 22 | over SSH               |
| HTTP 80             | → | HTTPS 443     | TLS                    |
| POP3 110 · IMAP 143 | → | 995 · 993     | implicit TLS           |
| LDAP 389            | → | LDAPS 636     | or StartTLS on 389     |
| SNMP v1/v2c         | → | SNMPv3        | adds auth + encryption |

## Port ranges

RFC 6335

| 0-1023                        | 1024-49151                                  | 49152-65535                    |
|-------------------------------|---------------------------------------------|--------------------------------|
| Well-known<br>system services | Registered<br>vendor apps                   | Dynamic<br>client source ports |
| TCP                           | Reliable, handshake: web, SSH, mail, SMB    |                                |
| UDP                           | Fast, no handshake: DNS, DHCP, NTP, VoIP    |                                |
| Both                          | DNS: TCP 53 for zone transfers, big replies |                                |

## What is listening on this host?

|                                                     |                   |
|-----------------------------------------------------|-------------------|
| <code>netstat -ano   findstr LISTENING</code>       | Windows           |
| <code>Get-NetTCPConnection -State Listen</code>     | PowerShell        |
| <code>ss -tulpn</code>                              | Linux             |
| <code>Test-NetConnection 192.0.2.10 -Port 22</code> | Remote port open? |
| <code>tasklist /FI "PID eq 4321"</code>             | Who owns the PID? |
| <code>lsof -i :443</code>                           | Who uses 443?     |

**CAUTION** `ss -p` and `lsof` need `sudo` to see other users' processes. Exit the root shell when done.

## Firewall rules of thumb

**DANGER** Never open these to the internet:

23 Telnet
135-139
445 SMB
3389 RDP
1433
3306
161 SNMP

|                |                                                   |
|----------------|---------------------------------------------------|
| <b>Default</b> | Deny all inbound, allow only what a service needs |
| <b>Remote</b>  | RDP/SSH through a VPN or gateway with MFA         |
| <b>Verify</b>  | A port is a hint, not proof: check the service    |
| <b>Log</b>     | Log denied traffic and review it weekly           |



- FTP: 21 = control, 20 = data (active mode)
- SNMP: 161 = manager polls, 162 = traps
- RADIUS = UDP, TACACS+ = TCP 49

- DHCP: server 67, client 68, both UDP
- Mail: 25 server-to-server, 587 client sends
- SSH 22 also carries SFTP and SCP